

CAT-2: Finance & Economics



CAT-2.1: Stock Market Trend Forecasting Dashboard

Forecasting and visualizing market trends from historical price data.

(Historical Prices) – Using decades of daily stock price data for all U.S. stocks and ETFs (a dataset on the order of ~15 million records of daily OHLCV – open, high, low, close, volume – across NYSE and NASDAQ), a team can build a comprehensive forecasting and analytics dashboard[3]. The project could have two components: (a) a predictive model that forecasts stock or index trends, and (b) an interactive visualization for financial insights. For the predictive model, students might select a subset of stocks or an index (like S&P 500) and implement time-series models to forecast future prices or volatility. This can include classical approaches (ARIMA, GARCH for volatility) and advanced ones like an LSTM or Prophet model, carefully validated by rolling time windows to avoid forward-looking bias. Emphasis should be placed on *back testing* – for example, simulating how a trading strategy based on their forecasts would perform, which teaches proper model evaluation in finance. The second component is a **financial data exploration tool**: using the rich dataset, the team can create a dashboard (in Python with Plotly/Bokeh, or a web app) that lets users select a stock and see long-term trends, compare volume spikes, or overlay events (earnings, news) on price charts. This requires strong data handling – potentially using a database or memory optimization to manage ~0.95 GB of data – and visual storytelling. An exciting extension is to incorporate **anomaly detection**: identify periods of unusual market activity (such as the 2008 crash or 2020 pandemic drop) and highlight how different sectors behaved. The final deliverable would not only predict price movements (with appropriate disclaimers about market randomness) but also provide an intuitive way to explore historical market behavior, giving students a taste of quantitative finance analysis.

✦ DATA

Important: Projects must rely on **publicly available, immediately accessible historical market data**. No paid financial databases (e.g., CRSP, Bloomberg, Refinitiv) or institutional subscriptions are permitted.

Teams proposing stock-market forecasting, back-testing, anomaly detection, or financial dashboards **must use one or more** of the following datasets.

1. Historical US Stock & ETF Prices (Kaggle Aggregated Datasets)

Primary dataset (recommended for scale + convenience)

- **Description:**
Large-scale historical OHLCV datasets covering U.S. equities and ETFs across NYSE and NASDAQ, commonly used for quantitative finance projects.
- **Contents:**
 - Daily Open / High / Low / Close / Volume
 - Thousands of tickers
 - Multi-decade coverage (often 1960s–present, depending on release)
- **Formats:**
CSV (single or partitioned by ticker)
- **Representative datasets (immediate download):**
 - 👉 <https://www.kaggle.com/datasets/jacksoncrow/stock-market-dataset>
 - 👉 <https://www.kaggle.com/datasets/borismarjanovic/price-volume-data-for-all-us-stocks-etfs>
- **Why this works well for DATA 501:**
 - Large enough (~10–20M rows) to require careful data engineering
 - Ideal for forecasting, back-testing, and dashboarding
 - No API rate limits or authentication friction

2. Stooq Historical Market Data (Direct CSV Access)

Reliable, no-friction alternative

- **Description:**
Free historical price data for U.S. stocks, indices, and ETFs provided by Stooq, widely used in academic and hobbyist finance projects.
- **Contents:**
 - Daily OHLCV prices
 - Major indices (S&P 500, Dow Jones, NASDAQ)
 - Individual equities and ETFs
- **Formats:**
CSV (downloadable per ticker or index)
- **Access (immediate):**
👉 <https://stooq.com/db/h/>
- **Best used for:**
 - Index-level forecasting
 - Long-horizon trend analysis
 - Lightweight data pipelines

3. NASDAQ Data Link (formerly Quandl) – Free Market Datasets

Structured financial time-series with metadata

- **Description:**
A curated collection of free financial datasets, including equity prices, indices, and macro-linked market indicators.
- **Contents:**
 - Stock and ETF prices (selected datasets)
 - Index time-series
 - Sector and economic indicators (depending on dataset)
- **Formats:**
CSV, API access
- **Access:**
 <https://data.nasdaq.com/search?filters=%5B%22Free%22%5D>
- **Note:**
Free tier datasets are sufficient for forecasting and dashboard development.

4. Alpha Vantage (Optional API-Based Source)

For students interested in live-data extensions

- **Description:**
Free API providing historical and near-real-time market data.
- **Contents:**
 - Daily and intraday OHLCV
 - Major indices and ETFs
 - Technical indicators
- **Formats:**
JSON, CSV via API
- **Access:**
 <https://www.alphavantage.co/>
- **Constraints:**
 - API rate limits apply
 - Best used for subsets or live-update demonstrations

 Methodological Requirement (Important)

Regardless of dataset choice, the **methodology must reflect sound financial data science practices**, including:

- Strict train/test separation using rolling or expanding windows
- Avoidance of forward-looking bias
- Explicit back-testing and performance evaluation
- Clear disclaimers regarding market unpredictability

Students are expected to demonstrate:

- Robust time-series modeling (ARIMA/GARCH/LSTM/Prophet, etc.)
- Scalable data handling for millions of rows
- Effective visual analytics for financial insight

Summary

- **Kaggle OHLCV datasets** provide the best balance of scale and ease
 - **Stooq** offers frictionless long-term historical access
 - **NASDAQ Data Link** adds structured, metadata-rich alternatives
 - All datasets are **open, free, and immediately accessible**
 - Suitable for forecasting, anomaly detection, back-testing, and dashboards
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Each table corresponds to one of the five Data Science pillars—(1) Mathematics, (2) Statistics, (3) AI/ML, (4) Visualization, and (5) Data Engineering, Ethics & Reproducibility—and outlines the pillar-specific foundations intended to provide a clear roadmap of techniques, approaches, methods, and tools for getting started on the project.

1 Mathematics (Foundations that actually get used)

Focus	Core topics and concepts
Time-series representation	• Price and return sequences • Log-returns vs raw prices
Trend and seasonality	• Long-term trends • Cycles and periodic effects
Volatility concepts	• Variance over time • Volatility clustering intuition
Forecast targets	• Price level vs return vs volatility
Scaling	• Normalization of prices and volumes
Purpose and relevance	Provides the mathematical grounding for modeling market behavior as evolving time-dependent signals

2 Statistics (Evaluation, realism, and defensibility)

Focus	Core topics and concepts
Descriptive statistics	• Summary statistics of returns and volume
Temporal data splitting	• Rolling or expanding train/test windows
Back-testing	• Simulated historical forecasting performance
Baseline models	• Random walk / no-change baselines
Evaluation metrics	• RMSE / MAE (forecast error) • Directional accuracy
Statistical sanity checks	• Stability across market regimes
Purpose and relevance	Ensures forecasts are evaluated realistically and without forward-looking bias

3 AI / ML (Time-series modeling)

Focus	Core topics and concepts
Classical models	• ARIMA for price/returns • GARCH for volatility
Machine learning	• Tree-based regression on lag features
Deep learning	• LSTM or sequence models (exploratory)
Feature construction	• Lagged returns • Rolling averages and indicators
Model comparison	• Classical vs ML vs deep models
Libraries	• statsmodels • scikit-learn • PyTorch / Keras (optional)
Purpose and relevance	Demonstrates progression from statistical forecasting to modern ML approaches

4 Visualization (Insight, not decoration)

Focus	Core topics and concepts
Price evolution	• Long-term price and index charts
Volume dynamics	• Volume spikes and liquidity patterns
Forecast visualization	• Predicted vs actual trajectories
Uncertainty	• Confidence bands or error ranges
Anomaly highlighting	• Market crashes or regime shifts
Interactive dashboards	• Stock selection and time-range controls
Tools	• Matplotlib • Plotly / Bokeh
Purpose and relevance	Turns raw financial time-series into interpretable trends and narratives

5 Data Engineering, Ethics & Reproducibility

Focus	Core topics and concepts
Data acquisition	• Public OHLCV datasets (Kaggle, Stooq)
Data organization	• Partitioning by ticker or date
Memory efficiency	• Chunked loading • Type optimization
Workflow	• Modular pipelines for modeling and visualization
Reproducibility	• Fixed train/test windows • Versioned datasets
Ethics & communication	• Clear disclaimers about market predictability
Documentation	• README describing assumptions and limitations
Final artifact	• Forecasting dashboard or analytical notebook
Purpose and relevance	Connects quantitative modeling with transparent, responsible financial analytics practice